

Rice Grain Classifier

Semester Project Plan

Computer Vision & Machine Learning

5-Member Team | 14-Week Semester | 2nd Year Project

1 What Are We Building?

We are building a smart app that can look at a photo of a rice grain and tell you:

1. What variety of rice it is (like Basmati, Jasmine, etc.)
2. How confident it is about its answer (like "96% sure it's Basmati")
3. Physical details about the grain -- its size, shape, and color

Think of it like Shazam, but for rice. You upload a photo, the app analyzes it, and gives you a full report.

Why Does This Matter?

Right now, rice grading is done by hand -- people sit and visually inspect thousands of grains. This is slow, inconsistent (different people may grade the same grain differently), and expensive. Our app does the same job in seconds, and it never gets tired or inconsistent. This has real-world value in the food and agriculture industry.

2 Our Dataset

We're using a free, publicly available dataset from Kaggle (a website where people share datasets for machine learning).

Rice Image Dataset

Attribute	Details
Name	Rice Image Dataset (by Murat Koklu)
Source	kaggle.com/datasets/muratkokludataset/rice-image-dataset
Total Images	75,000 photos of rice grains
Rice Varieties	5 types: Arborio, Basmati, Ipsala, Jasmine, Karacadag
Per Variety	15,000 images each (perfectly balanced)
Image Size	250 x 250 pixels, white background
Bonus File	106 pre-calculated features (size, shape, color)

What's in the 106 Features File?

This is a spreadsheet-like file where someone has already measured things about each grain for us:

Category	Count	What It Includes
Size	12 features	Area, perimeter, grain length, grain width, etc.
Shape	4 features	How round or elongated, aspect ratio, etc.
Color	90 features	Avg redness, greenness, blueness, brightness, etc.

- * Why is this great? We don't need to calculate these numbers ourselves -- they're already done! We can feed these numbers directly into our machine learning models.

3 The 3 Parameters We Analyze

When we analyze a rice grain, we look at 3 physical characteristics:

1. Size -- How big is the grain?

What we measure: Length (how long), Width (how wide), Area (how much space it covers). Measured in pixels. Why it matters: Basmati rice is long, while Arborio is short and fat.

2. Shape -- What's the grain's overall shape?

We calculate the length-to-width ratio (L/W ratio) and map it to a human-readable label:

L/W Ratio	Shape Label	Example Variety
More than 3.0	Long Slender	Basmati
2.1 to 3.0	Medium	Jasmine
1.1 to 2.0	Bold	Arborio
1.0 or less	Round	Karacadag

3. Color -- What color is the grain?

We measure average Red, Green, Blue (RGB) values from the image and label the color as White, Creamy, Brown, etc. Different rice varieties have slightly different shades.

Why Pixels and Not Millimeters?

Every image is 250 x 250 pixels, but we don't know how zoomed-in the camera was. Without that information, we can't convert to real-world millimeters. That's okay! For classification, what matters is the ratio between length and width, not the actual mm value. A Basmati grain will always be longer and thinner than Arborio regardless of measurement unit.

4 What Our App Will Show

When someone uploads a rice grain photo, they'll see a result card like this:



How It Works (5 Steps)

1. User uploads a rice grain photo
2. We clean the image (resize, fix brightness)
3. Our CNN model predicts the variety + confidence
4. OpenCV measures the grain's size, shape, and color
5. Everything is combined and shown in the result card

5 Our Tools

Think of these as the apps and software we need:

Tool	What It Is	What We Use It For
Python	Programming language	Everything is written in Python
Google Colab	Free online coding + GPU	Training AI models (no expensive PC needed)
TensorFlow	AI library by Google	Building deep learning models
Scikit-learn	ML library	Building SVM, Random Forest, KNN
OpenCV	Image processing library	Measuring size, shape, color from photos
Streamlit	Web app framework	Building the user interface
NumPy/Pandas	Data handling libraries	Working with numbers and tables
Matplotlib	Charting library	Making graphs for the paper

* Good news: Google Colab comes with most of these pre-installed. Just open a notebook and start coding!

6 The Models We'll Build

We're building several models and comparing them. This is what makes our project stand out.

What's a "Model"?

A model is a program trained on thousands of examples. After training, it can look at a new photo it has never seen and predict the rice variety. The more examples it sees during training, the better it gets.

Group A: Deep Learning (Learns from Images)

Model	What It Is	Expected Accuracy
Custom CNN	Neural network we design from scratch	~95-97%
MobileNetV2	Pre-built Google model, fine-tuned for rice	~99%

What's a CNN? It stands for Convolutional Neural Network. Imagine a system that looks at a photo in layers -- first it notices edges, then shapes, then patterns, and finally recognizes the object.

What's transfer learning? Instead of training from zero, we take a model already trained on millions of photos (MobileNetV2) and just teach it rice. Like hiring a photographer and teaching them to identify rice -- much faster than training someone from scratch.

Group B: Classical ML (Learns from Numbers)

These use the 106 pre-calculated features (numbers) instead of looking at images directly:

Model	How It Works (Simple Version)
SVM	Draws mathematical boundaries between rice types
Random Forest	Hundreds of yes/no decision trees, majority vote
KNN	Finds most similar grains and copies their label

! Why build both? In our paper, we compare them and answer: "Is deep learning actually better than simpler methods for rice classification?" This comparison is our paper's core contribution.

7 Ablation Study (Our Secret Weapon)

An ablation study means removing parts one at a time to see how much each matters. Like removing one ingredient from a recipe to see how the dish changes.

We train our best ML model using different parameter combinations:

Experiment	Features Used	# Features	Accuracy
Size only	Area, perimeter, axis lengths...	12	To be filled
Shape only	Eccentricity, extent, roundness...	4	To be filled
Color only	RGB, HSV values...	90	To be filled
Size + Shape	Combined	16	To be filled
Size + Color	Combined	102	To be filled
Shape + Color	Combined	94	To be filled
ALL THREE	Everything	106	To be filled

This answers: "Which physical property matters most when identifying rice?" Is it color? Shape? Size? Or all three?

- * This is very easy to do (just re-run the model with fewer columns) but looks extremely impressive in your paper. This is a genuine research contribution.

8 Team Roles (5 Members)

Each person owns a clear area. Everyone should understand the full project, but you're each responsible for delivering your part.

Member	Role	Responsibility
Person 1	Data & Preprocessing	Download dataset, explore it, split into train/test, image cleanup
Person 2	Deep Learning Engineer	Build Custom CNN + MobileNetV2, train on Colab, tune accuracy
Person 3	Classical ML Engineer	Build SVM, RF, KNN on 106 features, run ablation study
Person 4	App Developer	Build Streamlit web app, design result card, integrate OpenCV
Person 5	Paper Writer & Integrator	Write paper throughout, literature review, integrate code, presentations

How You'll Work Together

Person 1 prepares data first (everyone depends on clean data). Persons 2 and 3 work in parallel building different model types. Person 4 builds the app and plugs in trained models from Persons 2 & 3. Person 5 writes the paper throughout -- not just at the end -- and connects everything.

! Common mistake: Leaving the paper for the last week. Person 5 should start the literature review and methodology from Week 1. By the time models are trained, the paper should be 50% done.

9 Semester Timeline (14 Weeks)

Week	Phase	Tasks	Lead
Week 1-2	Research & Setup	Everyone: Set up. Person 3-4: research papers. Set up Google Colab, download dataset, create train/test split.	Everyone + Person 5
Week 3-4	Data Exploration	Person 1: Explore dataset (charts, statistics). Split into training (70%), validation (15%), test (15%).	Person 1
Week 5-7	Model Building	Person 2: Build Custom CNN + MobileNetV2. Person 3: SVM, RF, KNN. Person 5: Intro + Literature Review.	Persons 2, 3, 5
Week 8-9	Ablation Study	Person 3: ablation study. Person 2: Grad-CAM. Person 5: Methodology section.	Persons 2, 3, 5
Week 10-11	App Development	Person 4: Streamlit app (upload, classify, display). Person 1: OpenCV measurer.	Persons 4, 1
Week 12-13	Evaluation & Results	Person 2: Confusion matrices, accuracy tables, charts. Person 5: Results + Conclusion.	Everyone + Person 5
Week 14	Final Polish	Person 5: Polish everything. Clean code + GitHub. Finalize paper. Prepare presentation slides.	Everyone

10 Research Paper Outline

Suggested Title

"Automated Rice Grain Classification Using Deep Learning: A Comparative Study of CNN and Classical Machine Learning Approaches Based on Shape, Size, and Color Features"

Paper Sections

1. Abstract (~200 words)

Short summary of the entire paper -- problem, method, results, conclusion. Write this LAST.

2. Introduction (1-1.5 pages)

Why rice grading matters, problem with manual grading, what our project does, our contributions.

3. Related Work (1 page)

What other researchers have done and what gap our project fills.

4. Methodology (2-3 pages)

Dataset description, preprocessing, our 3 parameters, model architectures, ablation study design.

5. Results & Discussion (2 pages)

Model comparison table, ablation results, confusion matrices, Grad-CAM images.

6. Web Application (0.5 page)

Screenshot of Streamlit app, how a user interacts with it.

7. Conclusion & Future Work (0.5 page)

Summary, limitations (controlled dataset, no mm), future work (real-world images, mobile).

8. References

15-25 references from Google Scholar and IEEE Xplore.

Key Results Table (To Be Filled After Training)

Model	Accuracy	Precision	Recall	F1-Score
Custom CNN	--	--	--	--
MobileNetV2	--	--	--	--
SVM	--	--	--	--
Random Forest	--	--	--	--
KNN	--	--	--	--

11 Final Deliverables

- ☐ Trained Models -- Custom CNN + MobileNetV2 + SVM + Random Forest + KNN
- ☐ Working Web App -- Streamlit: upload grain photo, get variety + size + shape + color

- ❑ Ablation Study -- Table showing which parameters matter most
- ❑ Research Paper -- 6-8 page paper in proper academic format
- ❑ Presentation -- 10-12 slides with live demo of the app
- ❑ GitHub Repository -- Clean, organized code with README

12 What Will Impress Our Professor

What We Do	Why It's Impressive
Compare 5 different models	Shows we understand multiple approaches
Ablation study	Shows scientific thinking: 'what if we remove this?'
Working web app	Goes beyond a notebook -- it's a real product
Grad-CAM heatmaps	Shows we can explain WHY the AI decides
Paper written throughout	Shows discipline and research methodology
Honest about limitations	Professors respect intellectual honesty

13 Setup Guide

Step 1: Create a Kaggle Account

Go to kaggle.com and sign up for free.

Step 2: Open Google Colab

Go to colab.research.google.com -- free coding environment with GPU access.

Step 3: Download the Dataset (in Colab)

```
# Install Kaggle tool
!pip install kaggle

# Upload your Kaggle API key (kaggle.com > Settings > API)
from google.colab import files
files.upload() # Upload kaggle.json file

# Set up the key
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json

# Download rice dataset
!kaggle datasets download -d muratkokludataset/rice-image-dataset
!unzip rice-image-dataset.zip -d ./data/
```

Step 4: Save Work to Google Drive

```
from google.colab import drive
drive.mount('/content/drive')

# After training, save your model:
model.save('/content/drive/MyDrive/rice_model.keras')
```

! Important: Google Colab sessions expire after a few hours. Always save your trained model files to Google Drive so you don't lose your work!

14 Glossary -- Key Terms in Plain English

Term	Simple Explanation
CNN	AI that understands images by looking at patterns layer by layer
Transfer Learning	Taking a pre-trained AI and teaching it a new task
Training Data	Images we show the AI so it can learn (70% of dataset)
Validation Data	Images to check how AI learns during training (15%)
Test Data	Images AI has never seen -- used for final score (15%)
Accuracy	What percentage of predictions were correct
Precision	Of all grains model said were 'Basmati', how many actually were?
Recall	Of all actual Basmati grains, how many did model find?
F1-Score	Combined score of Precision and Recall
Confusion Matrix	Table showing which varieties get mixed up
Grad-CAM	Highlights which part of image the AI focused on
Ablation Study	Removing parts one at a time to see what matters
Epoch	One complete pass through all training images (we do 20-50)
Overfitting	When AI memorizes data instead of learning -- like cramming
OpenCV	Free Python library for image processing
Streamlit	Turns Python scripts into web apps with a few lines of code